

YEAR 7 MATHEMATICS

Chapter 1

Integers

WORKSHEETS

1

How to use this booklet

1 There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

2 Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

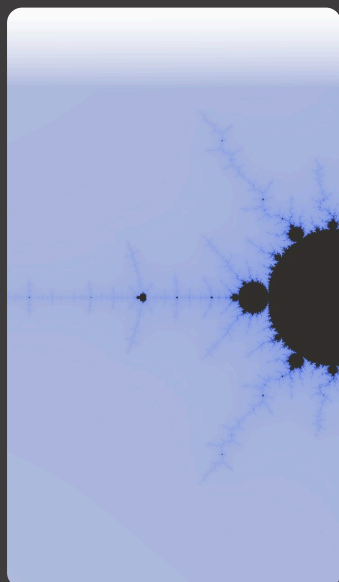
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
1.01	Integers and opposites	3		☐
1.02	The number line	4		☐
1.03	Comparing and ordering integers	5		☐
1.04	Adding integers	6		☐
1.05	Subtracting integers	7		☐
1.06	Multiplying integers	8		☐
1.07	Dividing integers	9		☐
1.08	Order of operations with integers	10		☐
1.09	Integers on a calculator	11		☐
1.10	Integer problems	12		☐

In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers

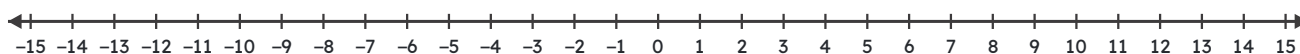


About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.



Easy

Questions 1–22

Write the opposite of each integer. It is the same distance from 0, on the other side.



WE DO With your teacher: copy the working.

E1 16

E2 7

YOU DO Your turn.

1 -59

59

2 41

3 9

4 47

5 -25

6 54

7 -14

8 -52

9 -43

10 23

11 27

12 -21

13 -9

14 29

Write an integer for each situation.



WE DO With your teacher: copy the working.

E3 63 floors up

YOU DO Your turn.

15 a fall of 8 points

-8

16 a fall of 48 points

17 50 m below sea level

18 52 minutes late

19 a gain of 4 kg

20 22 m below sea level

21 a loss of \$9

22 48 steps back

Medium

Questions 23–31

Simplify. The opposite of the opposite is the number itself.

WE DO With your teacher: copy the working.

E1 $-(-26)$

E2 $-(+10)$

YOU DO Your turn.

23 $-(-16)$

= the opposite of -16

= 16

24 $-(-36)$

= the opposite of 36

= _____

25 $-(-49)$

= the opposite of _____

= _____

26 $-(+7)$

27 $-(-57)$

28 $-(-2)$

29 $-(-25)$

30 $-(-26)$

31 $-(-29)$

Challenging

Questions 32–36

Answer each question. Write your reasoning on the lines.



WE DO With your teacher: copy the working.

E1 The opposite of a number is -16. What is the number?

E2 A diver is 27 m below sea level. Write this as an integer, then write its opposite.

E3 List all the integers whose distance from 0 is less than 4.

E4 If $n = -32$, find $-n$ and $-(-n)$.

YOU DO Your turn.

32 The opposite of a number is -34. What is the number?

opposite of ? = -34

the number is _____

33 A diver is 16 m below sea level. Write this as an integer, then write its opposite.

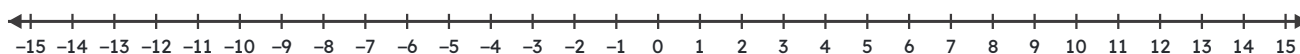
below sea level: _____

opposite: _____

34 List all the integers whose distance from 0 is less than 2.

35 If $n = -22$, find $-n$ and $-(-n)$.

36 The opposite of a number is -3. What is the number?



Easy

Questions 1–18

Which integer is further to the right on the number line? Write it on the line.

WE DO With your teacher: copy the working.

E1 5 or -4

E2 -2 or -3

YOU DO Your turn.

1 2 or -6

2

2 8 or -10

3 7 or -13

4 1 or -2

5 14 or -4

6 -2 or -11

7 -1 or 11

8 14 or 8

9 -13 or -7

10 3 or -12

11 10 or 1

12 3 or 2

13 15 or -5

14 -11 or 6

Start at the first number and move along the number line. Where do you land?

WE DO With your teacher: copy the working.

E3 start at -1, move 7 right

YOU DO Your turn.

15 start at -5, move 4 left

-9

16 start at 3, move 7 right

17 start at 5, move 2 left

18 start at 5, move 4 left

Medium

Questions 19–24

How many units apart are the two integers on the number line?

WE DO With your teacher: copy the working.

E1 -7 and 11

E2 -8 and 3

YOU DO Your turn.

19 3 and -1

-1 to 0 is 1, 0 to 3 is 3

1 + 3 = 4

20 11 and -12

-12 to 0 is 12, 0 to 11 is 11

_____ + _____ = _____

21 12 and -13

_____ to _____ is _____, _____ to _____

_____ + _____ = _____

22 5 and -11

23 -9 and 11

Which integer is exactly halfway between?

WE DO With your teacher: copy the working.

E3 -10 and -6

YOU DO Your turn.

24 -1 and 7

8 apart, half is 4

-1 + 4 = 3

Challenging

Questions 25–29

Count the integers or find the halfway point. Show your working.

WE DO With your teacher: copy the working.

E1 How many integers are between -29 and 8?

E2 Which integer is halfway between -24 and 0?

YOU DO Your turn.

25 How many integers are between -4 and 23?

-3 to -1: 3 integers, and 0

1 to 22: 22 integers

_____ + _____ + _____ = _____

26 Which integer is halfway between -15 and 15?

distance apart: _____

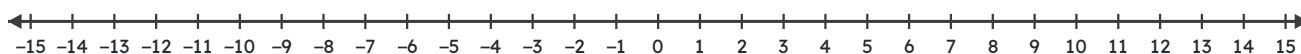
half: _____

_____ + _____ = _____

27 How many integers are between -24 and 7?

28 Which integer is halfway between -13 and 13?

29 How many integers are between -19 and 30?



Easy

Questions 1–22

Write $<$ or $>$ in the box. The integer further right is larger.

WE DO With your teacher: copy the working.

E1 $-7 \square -8$

E2 $-5 \square -10$

YOU DO Your turn.

1 $-1 \square -20$

2 $-12 \square 12$

3 $7 \square -11$

4 $2 \square 10$

5 $11 \square 0$

6 $-9 \square -6$

7 $-4 \square -6$

8 $-1 \square -17$

9 $-5 \square -15$

10 $-10 \square -6$

11 $-10 \square 0$

12 $-19 \square 3$

13 $1 \square 11$

14 $5 \square -10$

Write the smaller integer.

WE DO With your teacher: copy the working.

E3 -7 or -2

YOU DO Your turn.

15 2 or -14

16 0 or -14

17 7 or -12

18 -21 or -1

19 6 or -26

20 -24 or -21

21 -14 or 3

22 -30 or -28

Medium

Questions 23–28

Write the integers in ascending order (smallest first).

WE DO With your teacher: copy the working.

E1 $15, -8, -22, -9$

E2 $5, 14, -24, -18$

YOU DO Your turn.

23 $5, -20, 3, 0$

smallest: -20 , largest: 5

$-20, 0, 3, 5$

24 $-24, 5, 15, -6$

smallest: -24 , largest: 15

25 $10, -7, -8, 4$

smallest: _____, largest: _____

26 $-23, -14, 4, 15$

27 $6, -16, 20, -25$

Write the integers in descending order (largest first).

WE DO With your teacher: copy the working.

E3 $12, -27, 25, -32$

YOU DO Your turn.

28 $9, -22, -16, -13$

largest: 9 , smallest: -22

$9, -13, -16, -22$

Challenging

Questions 29–34

Simplify any $-(-n)$ first, then answer the question.

WE DO With your teacher: copy the working.

E1 Write three integers greater than -14 but less than -9 .

E2 Which is the smallest: $-(-7)$, -7 or $-(-12)$?

E3 Write in ascending order: $-(-2)$, -9 , -12 , 0

YOU DO Your turn.

29 Write three integers greater than -18 but less than -12 .

integers from -17 to -13

e.g. _____, _____, _____

30 Which is the smallest: $-(-7)$, -8 or $-(-9)$?

$-(-7) =$ _____, $-(-9) =$ _____

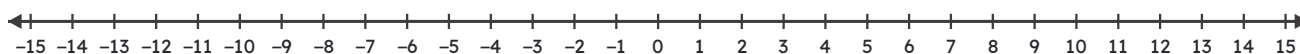
smallest: _____

31 Write in ascending order: $-(-4)$, -7 , -3 , 0

32 Write three integers greater than -19 but less than -11 .

33 Which is the smallest: $-(-2)$, -6 or $-(-11)$?

34 Write in ascending order: $-(-9)$, -7 , -11 , 0



Easy

Questions 1–22

Evaluate each sum. Start at the first number and move along the number line.

WE DO With your teacher: copy the working.

E1 $-4 + 8$ _____

E2 $-7 + 10$ _____

YOU DO Your turn.

1 $-7 + 1$ _____ **-6**

2 $-4 + 10$ _____

3 $-4 + 3$ _____

4 $-3 + 1$ _____

5 $-11 + 2$ _____

6 $-2 + 1$ _____

7 $-8 + 10$ _____

8 $-8 + 2$ _____

9 $-10 + 11$ _____

10 $-9 + 5$ _____

11 $-7 + 7$ _____

12 $-11 + 11$ _____

13 $-10 + 7$ _____

14 $-9 + 8$ _____

Evaluate. Adding a negative is the same as subtracting.

WE DO With your teacher: copy the working.

E3 $8 + (-1)$ _____

YOU DO Your turn.

15 $11 + (-1)$ _____ **10**

16 $-6 + (-7)$ _____

17 $10 + (-11)$ _____

18 $-4 + (-12)$ _____

19 $11 + (-4)$ _____

20 $12 + (-2)$ _____

21 $12 + (-3)$ _____

22 $7 + (-1)$ _____

Medium

Questions 23–31

Evaluate. Add from left to right.

WE DO With your teacher: copy the working.

E1 $-12 + (-11) + (-13)$ _____

E2 $-4 + (-3) + (-6)$ _____

YOU DO Your turn.

23 $-4 + 14 + (-5)$ _____

= $10 + (-5)$ _____

= 5 _____

24 $-7 + (-3) + 9$ _____

= $-10 + 9$ _____

= -1 _____

25 $10 + 2 + 5$ _____

= $12 + 5$ _____

= 17 _____

26 $-15 + 4 + 1$ _____

27 $3 + (-8) + (-5)$ _____

28 $-13 + 7 + 4$ _____

29 $13 + 9 + 2$ _____

30 $6 + 4 + (-3)$ _____

31 $4 + 6 + (-7)$ _____

Challenging

Questions 32–35

Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 The temperature is -7°C and rises 8°C . What is the new temperature? _____

E2 A lift at level -3 goes up 6 floors. Which level is it on? _____

E3 Sam has \$35 and spends \$51 on credit. What is the balance? _____

E4 Find the missing number: $-12 + \square = 6$ _____

YOU DO Your turn.

32 The temperature is -6°C and rises 16°C . What is the new temperature? _____

= $-6 + 16$ _____

= 10°C _____

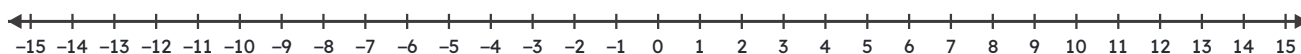
33 A lift at level -2 goes up 5 floors. Which level is it on? _____

= $-2 + 5$ _____

= level 3 _____

34 Sam has \$34 and spends \$62 on credit. What is the balance? _____

35 Find the missing number: $-5 + \square = 2$ _____



Easy

Questions 1–22

Evaluate. Subtracting moves you left on the number line.

WE DO With your teacher: copy the working.

E1 $-6 - 4$ _____

E2 $-5 - 4$ _____

YOU DO Your turn.

1 $-8 - 7$ -15

2 $-8 - 15$ _____

3 $2 - 2$ _____

4 $5 - 15$ _____

5 $5 - 13$ _____

6 $9 - 12$ _____

7 $6 - 10$ _____

8 $1 - 3$ _____

9 $-3 - 3$ _____

10 $0 - 8$ _____

11 $0 - 10$ _____

12 $8 - 2$ _____

13 $-8 - 10$ _____

14 $-6 - 3$ _____

Evaluate. Subtracting a negative is the same as adding.

WE DO With your teacher: copy the working.

E3 $12 - (-9)$ _____

YOU DO Your turn.

15 $-2 - (-11)$ 9

16 $10 - (-1)$ _____

17 $8 - (-8)$ _____

18 $2 - (-9)$ _____

19 $-12 - (-5)$ _____

20 $3 - (-3)$ _____

21 $11 - (-10)$ _____

22 $-4 - (-8)$ _____

Medium

Questions 23–28

Evaluate. Change each $- (-)$ to $+$, then add from left to right.

WE DO With your teacher: copy the working.

E1 $1 - (-15) + (-15)$ _____

E2 $7 - (-13) + (-6)$ _____

YOU DO Your turn.

23 $-13 - (-7) + (-7)$

$= -13 + 7 + (-7)$

$= -6 + (-7)$

$= -13$

24 $-5 - (-1) + 15$

$= -5 + 1 + 15$

$= -4 + 15$

$= \underline{\hspace{2cm}}$

25 $14 - (-10) + 11$

$= \underline{\hspace{2cm}} + \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$

$= \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$

$= \underline{\hspace{2cm}}$

26 $13 - (-7) + (-5)$ _____

27 $6 - (-9) + (-5)$ _____

28 $9 - (-7) + (-7)$ _____

Challenging

Questions 29–32

Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 The minimum temperature was -5°C and the maximum was 19°C . By how much did it rise?

E2 A submarine at -62 m rises 22 m . Where is it now?

E3 What is 2 less than -9 ?

E4 Find the difference between -6 m and 24 m .

YOU DO Your turn.

29 The minimum temperature was -10°C and the maximum was 11°C . By how much did it rise?

$11 - (-10)$

$= 11 + 10$

$= \underline{\hspace{2cm}}^{\circ}\text{C}$

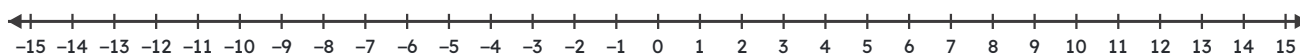
30 A submarine at -105 m rises 22 m . Where is it now?

$\underline{\hspace{2cm}} + \underline{\hspace{2cm}}$

$= \underline{\hspace{2cm}}\text{ m}$

31 What is 9 less than -4 ?

32 Find the difference between -15 m and 18 m .



Easy

Questions 1–22

Evaluate each quotient.
Same signs: positive.
Different signs: negative.

WE DO With your teacher: copy the working.

E1 $36 \div (-4)$ _____

E2 $-28 \div (-7)$ _____

YOU DO Your turn.

1 $8 \div (-8)$ -1

2 $90 \div (-9)$ _____

3 $-4 \div (-1)$ _____

4 $40 \div (-8)$ _____

5 $4 \div (-1)$ _____

6 $-40 \div (-5)$ _____

7 $-56 \div 7$ _____

8 $40 \div (-5)$ _____

9 $-30 \div (-10)$ _____

10 $-12 \div 3$ _____

11 $-4 \div (-4)$ _____

12 $-10 \div 1$ _____

13 $-24 \div 6$ _____

14 $15 \div (-5)$ _____

Find the missing number.

WE DO With your teacher: copy the working.

E3 $30 \div \square = -5$ _____

YOU DO Your turn.

15 $36 \div \square = 4$ 9

16 $8 \div \square = 8$ _____

17 $25 \div \square = 5$ _____

18 $-14 \div \square = 7$ _____

19 $-5 \div \square = 5$ _____

20 $-5 \div \square = -5$ _____

21 $-9 \div \square = 9$ _____

22 $-20 \div \square = 4$ _____

Medium

Questions 23–40

Evaluate. Divide from left to right.

WE DO With your teacher: copy the working.

E1 $8 \div 4 \div 2$ _____

E2 $-40 \div (-5) \div (-2)$ _____

YOU DO Your turn.

23 $-10 \div 1 \div 2$ _____

= $-10 \div 2$

= -5

24 $-9 \div 1 \div 3$ _____

= $-9 \div 3$

= -3

25 $8 \div (-1) \div (-4)$ _____

= $\square \div (\square)$

= $\square$

26 $-8 \div 2 \div 4$ _____

27 $-8 \div 2 \div (-2)$ _____

28 $-15 \div (-5) \div 3$ _____

29 $-50 \div (-5) \div (-2)$ _____

30 $-32 \div (-4) \div (-4)$ _____

31 $45 \div 5 \div 3$ _____

32 $10 \div 1 \div (-2)$ _____

33 $4 \div 4 \div 1$ _____

34 $32 \div (-4) \div 4$ _____

35 $16 \div 4 \div 2$ _____

36 $-60 \div 3 \div (-4)$ _____

37 $36 \div 3 \div 4$ _____

38 $6 \div (-2) \div 1$ _____

39 $24 \div (-3) \div (-4)$ _____

40 $-6 \div (-1) \div 3$ _____

Challenging

Questions 41–45

Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 A debt of \$160 is shared equally by 8 people. Write each share as an integer.

E2 The temperature fell 15°C over 3 hours. What was the change each hour?

E3 Find the missing number: $\square \div 8 = 4$

YOU DO Your turn.

41 A debt of \$114 is shared equally by 6 people. Write each share as an integer.

$-114 \div 6$

= $\square$ each

42 The temperature fell 12°C over 6 hours. What was the change each hour?

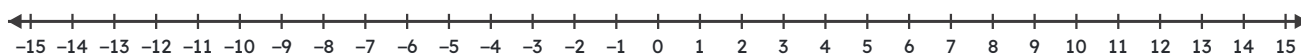
$\square \div \square$

= $\square^{\circ}\text{C}$ each hour

43 Find the missing number: $\square \div (-4) = -5$

44 A debt of \$126 is shared equally by 7 people. Write each share as an integer.

45 The temperature fell 24°C over 4 hours. What was the change each hour?



Easy

Questions 1–22

Evaluate. $\times$ and $\div$ before $+$ and $-$.



WE DO With your teacher: copy the working.

E1 $3 + (-6) \times (-6)$ _____

E2 $-9 - 5 \times (-6)$ _____

YOU DO Your turn.

1 $-5 - (-6) \times (-4)$ -29

2 $-12 - (-2) \times (-2)$ _____

3 $-6 - (-5) \times (-6)$ _____

4 $10 + 6 \times 4$ _____

5 $2 + 4 \times 5$ _____

6 $-3 - (-4) \times 4$ _____

7 $7 + (-6) \times (-2)$ _____

8 $9 - 2 \times (-2)$ _____

9 $3 - (-4) \times 4$ _____

10 $9 + 4 \times 2$ _____

11 $8 + 5 \times (-6)$ _____

12 $-10 + 2 \times (-6)$ _____

13 $-4 - 2 \times (-6)$ _____

14 $8 + (-1) \times (-3)$ _____

Evaluate. Brackets first.



WE DO With your teacher: copy the working.

E3 $(-6 - (-1)) \times 2$ _____

YOU DO Your turn.

15 $(2 + (-9)) \times 3$ -21

16 $(-5 - 7) \times 4$ _____

17 $(6 - 3) \times 3$ _____

18 $(2 + (-4)) \times 1$ _____

19 $(4 - (-9)) \times 1$ _____

20 $(7 - (-1)) \times (-4)$ _____

21 $(-5 - (-6)) \times 5$ _____

22 $(-6 + 4) \times (-2)$ _____

Medium

Questions 23–31

Evaluate. Do $\times$ and $\div$ first (left to right), then $+$ and $-$.

WE DO With your teacher: copy the working.

E1 $6 \times 1 + 2 \div 1$ _____

E2 $3 \times (-6) + (-16) \div 4$ _____

YOU DO Your turn.

23 $-5 \times (-6) - (-5) \div (-5)$
= $30 - 1$
= 29

24 $-1 \times (-2) + (-10) \div 2$
= $2 + (-5)$
= _____

25 $1 \times 6 + (-10) \div 2$
= _____ + (_____)
= _____

26 $1 \times 3 + 6 \div (-3)$ _____

27 $4 \times 1 + 15 \div 5$ _____

28 $6 \times (-4) - 8 \div 4$ _____

29 $6 \times (-2) - 20 \div 4$ _____

30 $-5 \times (-3) + 5 \div (-5)$ _____

31 $-6 \times 6 - 6 \div (-3)$ _____

Challenging

Questions 32–37

Evaluate. Brackets first, then $\times$ and $\div$, then $+$ and $-$. One step on each line.



WE DO With your teacher: copy the working.

E1 $7 - (-4 + 6) \times (-3)$ _____

E2 $6 - (-1 + (-2)) \times (-1)$ _____

YOU DO Your turn.

32 $-5 - (7 + 6) \times (-3)$
= $-5 - 13 \times (-3)$
= $-5 - (-39)$
= _____

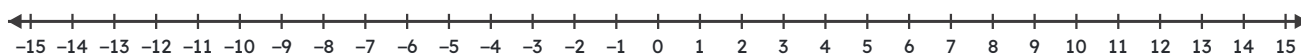
33 $5 - (2 + 8) \times (-1)$
= _____ - _____ $\times$ (_____)
= _____ - (_____)
= _____

34 $-5 - (5 + 8) \times (-3)$ _____

35 $2 - (7 + (-4)) \times (-4)$ _____

36 $7 - (8 + (-1)) \times 3$ _____

37 $-7 - (-4 + 5) \times (-3)$ _____



Easy

Questions 1–22

Use your calculator. Use the $(-)$ key for negative numbers.



WE DO With your teacher: copy the working.

E1 $252 + (-391)$ _____

E2 $-369 + 350$ _____

YOU DO Your turn.

1 $27 + 17$ 44

2 $-160 + (-124)$ _____

3 $24 + 106$ _____

4 $-198 + 77$ _____

5 $-269 + 255$ _____

6 $269 + (-274)$ _____

7 $267 + 146$ _____

8 $-158 + 224$ _____

9 $-396 + 241$ _____

10 $197 + (-106)$ _____

11 $59 + (-266)$ _____

12 $-303 + (-27)$ _____

13 $296 + 12$ _____

14 $224 + 100$ _____

Use your calculator.



WE DO With your teacher: copy the working.

E3 $43 \times (-4)$ _____

YOU DO Your turn.

15 22×3 66

16 $45 \times (-44)$ _____

17 $3 \times (-11)$ _____

18 $-45 \times (-20)$ _____

19 -22×12 _____

20 $17 \times (-33)$ _____

21 $31 \times (-32)$ _____

22 $13 \times (-33)$ _____

Medium

Questions 23–31

Use your calculator. Enter the brackets, then check the answer makes sense.

WE DO With your teacher: copy the working.

E1 $(-86 - (-81)) \times (-16)$ _____

E2 $(-3 - 67) \times (-11)$ _____

YOU DO Your turn.

23 $(-86 - (-8)) \times (-22)$

bracket: -78

$-78 \times (-22) = 1\,716$

24 $(12 - 22) \times 19$

bracket: -10

$_____ \times _____ = _____$

25 $(12 - (-59)) \times (-18)$

bracket: _____

$_____ \times (_____) = _____$

26 $(-44 - 62) \times (-21)$

27 $(45 - 80) \times (-4)$

28 $(-90 - 59) \times (-19)$

29 $(-31 - 52) \times (-15)$

30 $(57 - 83) \times 15$

31 $(-74 - (-90)) \times (-3)$

Challenging

Questions 32–38

Estimate first, then use your calculator. Is your answer close to the estimate?

WE DO With your teacher: copy the working.

E1 Estimate -60×44 , then calculate.

E2 Estimate -32×48 , then calculate.

YOU DO Your turn.

32 Estimate -34×17 , then calculate.

estimate: $-30 \times 20 \approx -600$

calculator: _____

33 Estimate -58×36 , then calculate.

estimate: $_____ \times _____ \approx _____$

calculator: _____

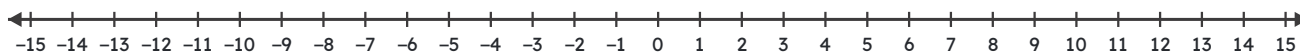
34 Estimate -42×35 , then calculate.

35 Estimate -46×28 , then calculate.

36 Estimate -38×30 , then calculate.

37 Estimate -57×13 , then calculate.

38 Estimate -48×26 , then calculate.



Easy

Questions 1–22

Find the new temperature.

WE DO

With your teacher: copy the working.

E1 5°C, rises 14°C

E2 14°C, falls 4°C

YOU DO

Your turn.

1 -4°C, falls 17°C

-21°C

2 -7°C, falls 16°C

3 -9°C, rises 15°C

4 11°C, falls 7°C

5 7°C, falls 5°C

6 -10°C, rises 10°C

7 5°C, falls 11°C

8 -2°C, falls 15°C

9 11°C, rises 10°C

10 12°C, falls 13°C

11 -2°C, falls 12°C

12 13°C, falls 17°C

13 -10°C, falls 14°C

14 13°C, rises 13°C

Find the new bank balance.

WE DO

With your teacher: copy the working.

E3 -\$72, deposit \$92

YOU DO

Your turn.

15 -\$9, withdraw \$103

-\$112

16 -\$23, deposit \$24

17 \$133, deposit \$70

18 \$118, deposit \$53

19 -\$73, withdraw \$18

20 \$103, withdraw \$78

21 -\$9, deposit \$86

22 \$13, deposit \$42

Medium

Questions 23–29

Find the difference between the two heights. Write a number sentence.

WE DO

With your teacher: copy the working.

E1 a hill 267 m above sea level and a reef 57 m below

E2 a hill 280 m above sea level and a reef 74 m below

YOU DO

Your turn.

23 a hill 106 m above sea level and a reef 10 m below

106 - (-10)

= 116 m

24 a hill 66 m above sea level and a reef 19 m below

66 - (-19)

= ____ m

25 a hill 52 m above sea level and a reef 32 m below

____ - (____)

= ____ m

26 a hill 174 m above sea level and a reef 48 m below

27 a hill 388 m above sea level and a reef 57 m below

28 a hill 129 m above sea level and a reef 45 m below

29 a hill 28 m above sea level and a reef 88 m below

Challenging

Questions 30–32

Write a number sentence for each step, then answer the question.

WE DO

With your teacher: copy the working.

E1 A diver is at -45 m. She rises 4 m a minute for 4 minutes. Where is she now?

E2 A game scores +4 for a win and -3 for a loss. Mia wins 3 and loses 5. What is her score?

E3 It is -2°C at 6 a.m. It warms 3°C an hour for 3 hours. What is the temperature?

YOU DO

Your turn.

30 A diver is at -30 m. She rises 5 m a minute for 5 minutes. Where is she now?

5 × 5 = 25

-30 + 25

= ____ m

31 A game scores +4 for a win and -1 for a loss. Mia wins 4 and loses 4. What is her score?

____ × ____ = ____

____ × (____) = ____

____ + (____) = ____

32 It is -3°C at 6 a.m. It warms 5°C an hour for 6 hours. What is the temperature?